

Fluoride Metabolism



Objectives

- At the end of the lecture, students would be able to:
 - ✓ Know the sources and daily requirement of fluoride.
 - ✓ Explains the functions in detail.
 - ✓ Describe the term “fluorosis”.

FLUORIDE (F⁻)

- Fluoride is the negative ion of the element fluorine.
- It's widely found in nature, in trace amounts.
- It occurs naturally in air, soil, plants, rocks, fresh water, seawater, and many foods.

Sources:

- Fluoride is derived in human from drinking water.
- Other sources: Tea and salmon contain small amount of fluoride.

Fluoride vs Fluorine

Fluoride is either the fluorine ion or a compound containing fluorine.

F⁻
fluoride ion

NaF
sodium fluoride

Fluorine is an element on the periodic table.

9
F
Fluorine
18.9984



sciencesnotes.org

Requirements:

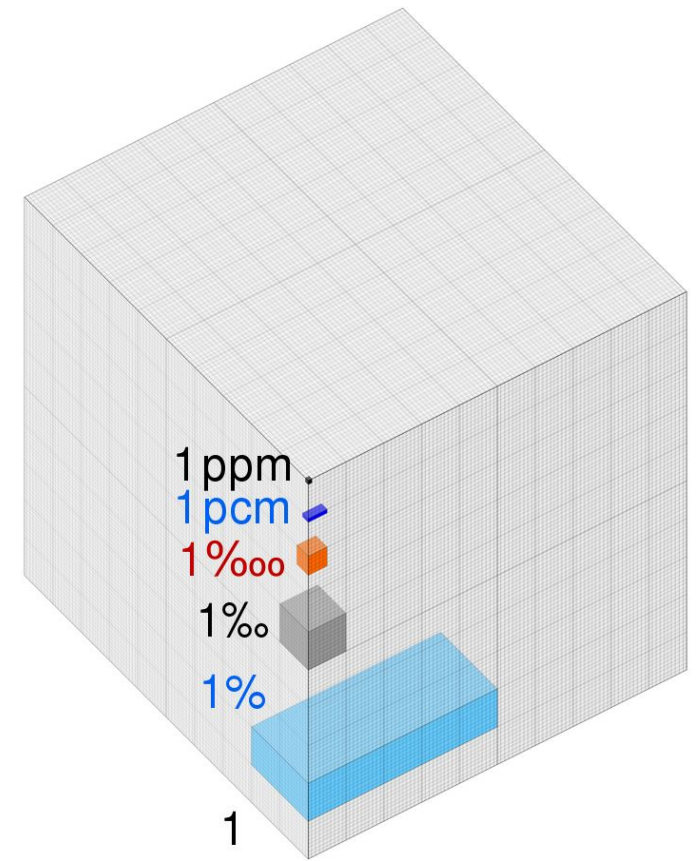
- About one part of fluorine in one million parts of drinking water (one PPM) seems to serve the daily requirement of fluorine in human adults and children.
- Daily intake of fluoride should not exceed 3 mg as it is a toxic element.
- For an adult individual, the lethal dose is 2.5 gm.

FUNCTIONS

- Fluoride is present in trace amounts in all mineralized tissues of the body such as enamel, dentin, and bone and plays an important role in the mineralization of bones and teeth, helps in tooth development, normal maintenance and hardening of dental enamel and prevention of “dental caries”.
- In mineralized tissues and biomaterials, fluoride ions increase the stability of mineralized tissues and materials by decreasing the solubility of hydroxy-apatite mineral.

- The protective effects of fluoride on dental health were first observed in 1930 as there was less tooth decay in communities consuming naturally fluoridated water compared to non-fluoridated areas.
- Due to these beneficial effects of fluoride, it was introduced into dentistry in 1940 and since then, it is being added to various consumer products.

- Greater than 1.2 PPM of fluorine in drinking water of infants/children may increase fluoride contents of the enamel and dentine, may reduce Ca deposition in those tissues and may cause mottling of enamel, in newly erupted permanent teeth: Discoloration, corrosion and stratification of enamel.
- It also promotes normal bone development, increases retention of Ca^{++} and PO_4 and prevent old age osteoporosis.
- High fluoride intakes may raise the fluoride content of bone, stimulate osteoblast activity and cause an abnormal rise in calcium deposition and increased density of bone.



Mottling of teeth

Fluoride

Fluoride strengthens teeth and reduces your risk of cavities. Specifically, fluoride:



Remineralizes your tooth enamel.



Reverses early tooth decay.



Slows down the process of demineralization.



Prevents the growth of cavity-causing bacteria.

Fluoride Toxicity “Fluorosis”:

- Excess of fluoride in water or diet is harmful and is considered to be the main cause of the crippling disease known as ‘fluorosis’.
- The fluoride levels in GI tract, blood and urine have been reported to be high in fluoride toxicity.
- High fluoride concentration inhibits particularly Mg^{++} dependent enzymes. The structural and biochemical changes led to muscle wasting and impairment of energy metabolism.
- 10 to 46% reduction in protein contents of various organs, e.g. adrenal gland, cardiac muscles, kidneys, lungs, pancreas, skeletal muscles, spleen, stomach, testes and spinal cord, suggestive of inhibition of protein synthesis by fluorides.

cont..

- Steroid production was impaired because of depletion of β -hydroxysteroid dehydrogenase enzyme.
- Collagen content is found to be reduced and its biosynthesis adversely affected. Formation of deficient collagen fibres with abnormal biochemical sites, provide an impetus for pathological calcification which occurs during fluoride intoxication.

THANK YOU